THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. What is the sum of the first 20 natural numbers $1 + 2 + 3 + \cdots + 20$? 1

(a) 210 **(b)** 420 **(c)** 230 **(d)** 400

2. You face North and turn clockwise through 135 degrees. What further clockwise turn (in degrees) brings you back to facing North? 1

(a) 315 **(b)** 225 **(c)** 45 **(d)** 135

3. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 68: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

4. Using the table, how many more are there of Mango than Guava? 1

Item	Number
Guava	25
Mango	45
Apple	30

(a) 15 **(b)** 25 **(c)** 70 **(d)** 20

5. The number of prime numbers between 20 and 50 is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 9

6. The area of a rectangle is 150 cm^2 and its length is 25 cm. Its breadth (in cm) is: 1

(a) 12 **(b)** 6 **(c)** 31 **(d)** 19

7. $2\frac{1}{4} + 1\frac{1}{2} =$ 1

(a) $3\frac{1}{4}$ **(b)** $3\frac{1}{8}$ **(c)** $3\frac{3}{4}$ **(d)** $3\frac{1}{3}$

8. The diagonals of a rectangle are each 16 cm long and meet at O. The length OA (cm) is: 1

(a) 16 **(b)** 9 **(c)** 7 **(d)** 8

9. A figure coincides with itself after turns of 72, 144, 216, ... degrees, up to a full turn of 360°. Its order of rotational symmetry is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

10. The number of integers between -8 and 2 is: 1

(a) 8 **(b)** 11 **(c)** 9 **(d)** 10

11. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 17th one is: 1

(a) 306 **(b)** 34 **(c)** 272 **(d)** 289

12. 5 rays start from a point O, no two of them in the same direction. If every pair of rays makes one angle, how many angles are there? 1

(a) 10 **(b)** 4 **(c)** 5 **(d)** 20

13. The digit sum of 98765 is: 1

(a) 44 **(b)** 35 **(c)** 26 **(d)** 40

14. The total number of students in the table is: 1

Sport	Students
Cricket	35
Tennis	65
Football	50

(a) 160 **(b)** 115 **(c)** 145 **(d)** 150

15. The prime factorisation of 120 has how many prime factors (counting repeats)? 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

16. Fencing a rectangular plot of perimeter 120 m costs ₹20 per metre. The total cost is (₹): 1

(a) 2500 **(b)** 2380 **(c)** 2420 **(d)** 2400

17. Arrange in ascending order: $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{12}$. 1

(a) $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{12}$ **(b)** $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{12}$
(c) $\frac{3}{8}$, $\frac{5}{12}$, $\frac{1}{2}$ **(d)** $\frac{1}{2}$, $\frac{5}{12}$, $\frac{3}{8}$

18. In rectangle ABCD, diagonals meet at O and $\angle AOB = 80^\circ$. Then $\angle OAB$ is (degrees): 1

(a) 40 **(b)** 50 **(c)** 100 **(d)** 60

19. A regular polygon has 9 lines of symmetry. The number of its sides is: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 11

20. A diver at -24 m (below sea level) swims up 10 m. His new position (m) is: 1

(a) -16 **(b)** -14 **(c)** -12 **(d)** 6

21. In the pattern $1, 3, 6, 10, \ldots$ what is the 12th term? 1

(a) 66 **(b)** 78 **(c)** 91 **(d)** 144

22. A protractor shows the arm of an angle at 55° on the inner scale. The reading on the outer scale for the same arm is: 1

(a) 125 **(b)** 35 **(c)** 55 **(d)** 305

23. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 25 and 47, the number above them is: 1

(a) 82 **(b)** 1175 **(c)** 22 **(d)** 72

24. The number which occurs most often in the data 2, 4, 5, 3, 2, 2, 3, 2, 1, 4, 4, 5 is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

25. In the Sieve of Eratosthenes up to 50, we need to cross out multiples of primes up to: 1

(a) 11 **(b)** 25 **(c)** 5 **(d)** 7

26. How many square tiles of side 30 cm are needed to cover a floor of 600 cm by 450 cm? 1

(a) 9000 **(b)** 35 **(c)** 300 **(d)** 70

27. A pizza is cut into 10 equal slices. Riya eats 6 slices. The fraction she eats in simplest form has denominator: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 4

28. A circle of radius 5 cm and a concentric circle of radius 9 cm are drawn. The width of the ring between them (cm): 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

29. Which of these letters has both a vertical and horizontal line of symmetry? 1

(a) S **(b)** N **(c)** A **(d)** H

30. The absolute value of any non-zero integer is: 1

(a) equal to its opposite **(b)** negative
(c) zero **(d)** positive

31. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 216 is: 1

(a) 265 **(b)** 343 **(c)** 234 **(d)** 49

32. An angle is formed when two rays meet at a common: 1

(a) midpoint **(b)** end point of a segment
(c) centre **(d)** initial point (vertex)

33. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 7, 2, 5. The top number is: 1

(a) 18 **(b)** 16 **(c)** 14 **(d)** 28

34. In the data 4, 5, 5, 5, 5, 5, 3, 1, 5, 5, 4, 2, how many values are greater than 4? 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 3

35. The smallest prime number that is a factor of 143 is: 1

(a) 143 **(b)** 11 **(c)** 13 **(d)** 1

36. How many centimetres are there in 7 m? 1

(a) 710 **(b)** 700 **(c)** 690 **(d)** 1700

37. 3 chapatis are shared equally among 5 children. Each child gets (of a chapati): 1

(a) $\frac{3}{5}$ **(b)** $1\frac{2}{3}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{2}{5}$

38. The diagonals of a rectangle that is not a square: 1

(a) are equal but not perpendicular **(b)** are unequal
(c) are parallel **(d)** are perpendicular

39. The smallest angle of rotation of a figure is 60°. Its order of rotational symmetry is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 8

40. A person has a debt of ₹75 and then earns ₹50. The balance (₹) is: 1

(a) -15 **(b)** -27 **(c)** -25 **(d)** -23

41. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 9th term is: 1

(a) 89 **(b)** 55 **(c)** 76 **(d)** 34

42. What is the smaller angle between the hands of a clock at 9 o'clock? 1

(a) 120 **(b)** 90 **(c)** 270 **(d)** 60

43. Add 36 and its reverse 63. The sum is: 1

(a) 88 **(b)** 99 **(c)** 121 **(d)** 110

44. A pictograph where one symbol = 20 shows 3 full symbols and a half symbol. The number shown is: 1

(a) 60 **(b)** 70 **(c)** 61 **(d)** 80

45. Which of the following is divisible by 6? 1

(a) 4219 **(b)** 4221 **(c)** 4218 **(d)** 4220

46. The area of a triangle with base 18 cm and height 7 cm is (in cm^2): 1

(a) 45 **(b)** 63 **(c)** 81 **(d)** 70

47. $\frac{3}{4} + \frac{1}{6} =$ 1

(a) $\frac{2}{5}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{11}{12}$

48. A square of side 6 cm: the length of the segment joining mid-points of two opposite sides (cm) is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 9

49. A kite has how many lines of symmetry? 1

(a) 1 **(b)** 0 **(c)** 2 **(d)** 4

50. 15 + (-22) = 1

(a) 37 **(b)** -7 **(c)** -5 **(d)** -9

END OF QUESTION PAPER **Total: 50 Marks**

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